Do Now · retrieval
Linked lists and pointers

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Push 2, 5, 8 then pop. What remains bottom-to-top?

Q2

What is the first index of a C# array?

Q3

What is a base case?

Q4

Queue A,B,C: dequeue once and enqueue D. What remains?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Follow references — head = 0 means the first node has value 10. next[0] initially points at node 1 (20).

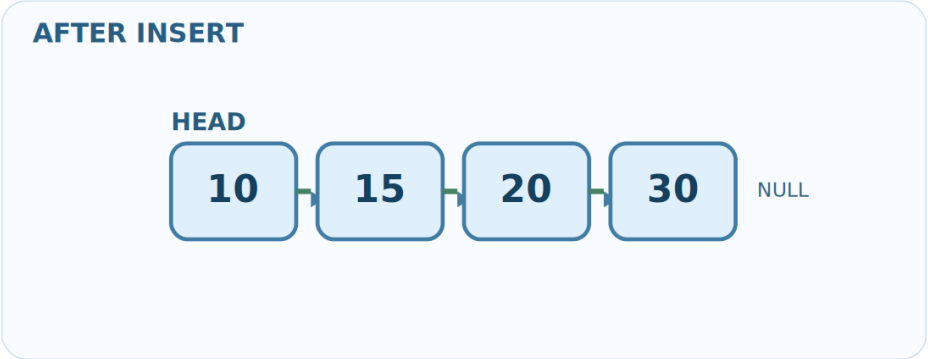
Core idea 2

Prepare the new node — The new node at index 3 stores 15 and next[3] = 1, preserving the rest of the list.

Core idea 3

Change the existing link — Set next[0] = 3, producing the chain 10 → 15 → 20 → 30.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

node — A list element containing data and a link.

next pointer — Reference or index of the next node.

head — Reference to the first list node.

null sentinel — Marker such as −1 in an index implementation meaning no next node.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

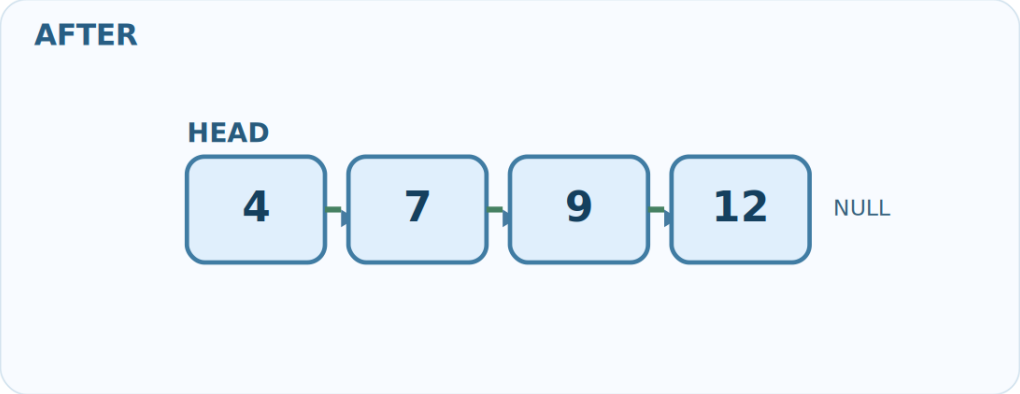
Problem

Insert value 7 between 4 and 9 in the linked list 4 → 9 → 12.

Follow the worked method

1. Predict — Create node 7 and set its next link to the existing node 9.
2. Confirm and explain — Point node 4 at node 7. The chain becomes 4 → 7 → 9 → 12.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

First link new node 6 to node 8.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

In linked list 5 → 10 → 15, what follows node 10?

Question 2

A node with value 12 must be inserted between 7 and 20. State the two link changes.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Why does inserting into a linked list avoid shifting array elements but may still take O(n) time?

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A singly linked list always supports direct O(1) access to its twentieth element.

Use key vocabulary from memory

1. Explain node in your own words.

2. Explain next pointer in your own words.

3. Explain head in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.